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SUMMARY

I am a Software Engineer, with a strong passion for GNU/Linux, Unix systems, and Systems Programming, and a deep interest in their evolution from inception to the present. I am seeking a role in **Linux/Unix Systems Engineering or Programming** to contribute my skills while gaining hands-on experience in real-world projects.

EXPERIENCE

Platform Software Engineer *Sonnen GmbH*

December 2025 - Now

- Migrated multiple legacy cron jobs to **systemd** services and timers across sonnen's embedded Linux system, that integrate properly into **dpkg** lifecycle hooks, which improves logging and troubleshooting and centralizes the system management from systemd.
- Improved various **NetworkManager dispatcher scripts** to properly update and configure the system with **exponential backoff strategy**.
- Implemented a **SoftAP** feature on sonnen's embedded Linux, bringing up a virtual access point over WiFi by orchestrating interface setup, dynamic config rendering for **hostapd** and **dnsmasq**, and iptables firewall rules — all managed through systemd units and a CLI tool.

SKILLS & LANGUAGES

Technical skills & Tools:

- C, C++, Python, Java, OOP, and Data structures
- GNU/Linux, Bash, and docker
- Linux Network
- Git, GitHub, vim, and tmux

Languages:

- Arabic: Native proficiency
- English: Intermediate proficiency (B1)

PROJECTS

c-toolkit

July 2025 - September 2025

- Developed reusable C99-based utilities, including:
 - **Arena allocator**: Developed thread-safe, region-based, and page-aligned memory allocator using **mmap**
 - **String**: Developed String utilities inspired by **Glib** from **Gnome Project**

BigNum

April 2025 - June 2025

- Developed a C99 arbitrary-precision arithmetic library for cryptographic applications with architecture-dependent 32/64-bit word representation using arena allocation for memory management.
- Designed a robust API by exploring **OpenSSL** and **libtommath** libraries, and implemented core big number algorithms such as Euclidean division and modular inverse.

imglib

November 2024 - January 2025

- Developed a lightweight, portable C library for image manipulation on Unix-like systems.
- Followed the **Suckless** C coding style and wrote a portable Makefile, avoiding GNU-specific extensions.
- Implemented robust error handling system inspired by **ffmpeg** and stride-based indexing for efficient pixel access.

Linux Kernel With lockdep

April 2024

- Configured and built the Linux kernel to enable lockdep module which helps in debugging deadlocks in kernel space, then booted into the custom kernel to verify lockdep functionality.

CONTRIBUTIONS

- **Unikraft:**
 - Fixed documentation issues in the main **README** and **sub-READMEs** in the examples directory for **unikraft/catalog** repo [[#231](#)] [[#232](#)]

EDUCATION

Faculty of Computers and Artificial Intelligence at Cairo University.

- Bachelor’s Degree in Information Technology

October 2021 - August 2025

CERTIFICATES

- CCNA: Introduction to Networks

July 2024 - August 2024
- CCNA: Switching, Routing, and Wireless Essentials

July 2024 - August 2024

EXTRACURRICULARS

IEEE Cairo University Student Branch

Networks Instructor

- Contributed to the development of technical content for the committee sessions and instructed real sessions for network committee members

March 2024 - June 2024